

Q17 Ans: C

Force on a charge in the field is given by $F = qE$. Since it's a uniform electric field, E is constant. Both proton and electron have the same magnitude of charge q .

Therefore the proton should experience the same magnitude of force as the electron.

The other options are correct statements because

Option A: Since the work done by the electric force on both the electron and proton is positive, therefore change in electric potential energy is negative for both ($W_{\text{by electric force}} = -\Delta U$)

Hence they both lose electric potential energy.

Option B: The proton experiences an upward electric force that is of same magnitude as that on the electron but has a larger mass.



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Therefore it will experience a smaller acceleration and undergo a smaller deflection.
Hence the work done by electric force is smaller for proton, giving it a smaller $|\Delta U|$.

Option D: The proton loses less electric potential energy than the electron. By conservation of energy, the gain in kinetic energy must be less than that of the electron.

Q18 Ans: D